THE UNIVERSITY OF SYDNEY

MTRX3760 - Lab 3
Containers

This assignment contributes 5% towards your final mark. It is to be completed individually, not in groups. Total Marks: 100.

This assignment is due before the start of your allocated lab session in one week, i.e. before the start of the Week 5 lab session for Thurs/Fri labs, and before the start of the Week 6 lab session for Monday labs. Late assignments will be subjected to the University's late submission policy unless accompanied by a valid special consideration form. Plagiarism will be dealt with in accordance with the University of Sydney plagiarism policy. **Lab submissions will not be accepted more than a week after the due date.**

Assignments will be assessed based on the following components. Incomplete submissions will have severely reduced marks:

- **Report**: Submit a .pdf report using the class Canvas site. The report can be very simple, and needs only directly address the questions laid out in the assignment.

  The cover page for your report should include your SID and tutorial section, but do not include your name to comply with the University's anonymous marking policy.

  Code appendix: The appendix of your report **must** contain a **printout** (in text format) of all source code written for the assignment. File header comments and proper formatting will be critical to making this section readable. Unintelligible code attachments will result in loss of marks.

- **Code**: Submit your code (code only, no binaries) in a .zip file via the canvas site.

Please see Lab 1 for hints on preparing your report, including how to format code for inclusion in the appendix.

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Occupancy Map

For this question, do all your development under revision control using git. Part of the question will ask for your git commit history in a specific format, and you will be assessed based on the quality of that log, showing appropriate use of revision control practices in your development process.

Starting from the provided code in `Lab3_MainQuestion_Code.zip`, you will design, program, and compare the performance of two alternative approaches to storing occupancy grid maps. Example data are provided in the zip file, along with an abstract base class that includes an evaluation function to validate and time the performance of your approaches.

The base class knows how to read the test data and evaluate your approaches. Please refer to the provided code to see how the base class works, and for example usage of the evaluation function.

The two approaches you design should use a **hash table** and **2D vector (vector of vectors)** data structures, respectively. You may employ the C++ Standard Library implementations of these data structures.

Your program should evaluate both your approaches using the base class evaluation function. An example output for evaluating a single approach is shown below:

```
Evaluating Hash table-based approach using test points from
ExampleObservations_Small.txt and ExampleNotObserved_Small.txt
Loading the file of observations... ExampleObservations_Small.txt
PopulateForEval loaded 924 observed test points
Loading the file of test points that don't correspond to observations...
ExampleNotObserved_Small.txt
PopulateForEval loaded 912 unobserved test points
Measuring runtime...
Total time to test all points: 0.737651 ms
```

[30 Marks] **Functionality and understanding:** In your report show the output of your program, including the performance of both of your approaches. This and the code in your attached .zip file will be assessed for **correct functionality**.

Please also address the following

- In your report, succinctly **explain** why one was **faster** than the other.
- In your report **estimate** how much **memory** each method required to store its map.
- In your report, **estimate** how much **memory** would be required by each method if observations could take values up to 1 million instead of 2047.



[30 Marks] **Code Quality and Development Process**: Include in your report your git commit history as reported by:
`git log --oneline`
note: copy/paste the text, screenshots are unacceptable
Your code and commit history will be assessed based on **style** and **coding and development best practices** as taught in lecture.

[40 Marks] **Design**: In your report provide a UML class diagram for your design, excluding member variables and functions, i.e. showing the overall structure of the program. This diagram and your code will be assessed based on appropriate use of the **design principles** taught in lecture.